INCH-POUND

MIL-DTL-3866H
13 December 2005
SUPERSEDING
MIL-G-3866G
17 June 1987

DETAIL SPECIFICATION

# GLOVES, CLOTH, COTTON, KNITTED, LIGHTWEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers lightweight, knitted, cotton gloves.

1.2 Classification. The gloves will be of the following types and sizes as specified (see 6.2):

Type I – Film Handlers
Type II – Inserts

Schedules of Sizes

Small
Medium

**Comments, suggestions, or questions on this document should be addressed to (Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CNR, 700 Robbins Ave., Philadelphia, PA 19111-5096) or emailed to James.Buckley@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.**

AMSC N/A FSC 8415

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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks from a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

**SPECIFICATIONS**

FEDERAL

| | |
|---|---|
| A-A-52094 | - Thread, Cotton |

MILITARY

| | |
|---|---|
| MIL-DTL-32075 | - Label: For Clothing, Equipage and Tentage, (General Use) |

(Unless otherwise indicated, copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.docs.dla.mil)

* 2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitations (see 6.2).



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ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM-D-3776 – | Mass per unit area (weight) of fabric |
| ASTM-D-3887 – | Standard specification for tolerances for knitted fabrics |
| ASTM-D-3787 – | Bursting strength of knitted goods consistent rate of traverse (CRT) ball test |
| ASTM-D-629 – | Quantitative analysis of textiles |
| ASTM-D-6193 – | Stitches, seams and stitchings |

(Application for copies should be addressed to, ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQC Z1.4 Sampling Procedures And Tables For Inspection By Attributes

(Applications for copies should be addressed to; ANSI/ASQC, American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

* 2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document will take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.2).

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this document may appear in the sample in which case this document shall govern.

3.3 Material. (See 6.4).

3.3.1. Fabric. The fabric for the gloves shall be a plain (jersey) circular knit fabric made with singles cotton yarn. The finished fabric shall be scoured and bleached. The use of optical bleaches (brighteners) is prohibited. The finished fabric shall conform to the following physical requirements when tested as specified in 4.2.1.1:

| | | |
|---|---|---|
| a. Weight per square yard, minimum | - | 3.4 ounces |
| b. Wales per inch, minimum | - | 28 |
| c. Courses per inch, minimum | - | 26 |
| d. Bursting strength, minimum | - | 50 pounds |



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3.3.1.1.Nonfibrous material content. The total sizing, finishing, and other nonfibrous material content of the finished fabric shall not exceed 2.0 percent when tested as specified in 4.2.1.1.

* 3.3.2. Thread. The cotton thread for seaming and stitching shall be white and shall conform to type IA2, ticket No. 50, 3 ply and ticket No. 70, 2 ply of A-A-52094.

* 3.4 Label. Each glove shall be marked with a combination identification and size label conforming to type III or IV, class 4 of MIL-DTL-32075, except the size shall be indicated on the bottom line rather than vertically. The label shall withstand fastness to laundering as specified in MIL-DTL-32075.

3.5 Design. The glove shall be an ambidextrous design with an unfinished wrist without cuff as shown on Figure 1. Type I gloves shall be inseam constructed and Type II gloves shall be outseam constructed.

3.6 Patterns. Standard patterns will be furnished by the Government. The standard patterns shall not be altered in any way and are to be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns. The patterns for Type I and Type II gloves shall be identical. The pattern parts and cutting quantities shall be as specified in Table I.

TABLE I. Pattern parts

| Parts | Cutting quantity per pair |
|---|---|
| Back | 2 |
| Front | 2 |

3.7 Construction.

3.7.1 Manufacturing operations requirements. The gloves shall be inseam or outseam sewn, as applicable, and shall be made by and with the use of all operations listed in Table II. The contractor is not required to follow the exact sequence of operations.

* 3.7.2 Stitches, seams, and stitchings. All stitches, seams, and stitching shall conform to ASTM-D-6193. The type of seam, stitching, and stitches per inch shall be as specified in Table II. When two or more methods, seams, or stitches are given for the same part of an operation, any one of them may be used. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur.

3.7.2.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.



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3.7.2.2 Type 301 stitching. Ends of 301 stitching shall be backstitched or overstitched 1/4 inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so there will be no loose stitching resulting in a loose bobbin or top thread or no excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the sewn materials.

3.7.2.3 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching.

3.7.2.4 Repairs of stitching. Repairs of stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

TABLE II. Manufacturing operations requirements

| No. Requirements | Stitch Type | Seam and Stitching Type | Stitches per Inch | Thread | |
|---|---|---|---|---|---|
| | | | | Needle | Bobbin/Looper |
| 1. Cutting. The gloves shall be cut in accordance with the patterns. There are two parts to each glove, a front and a back. All component parts shall be cut from the same piece of material with the length direction (cuff to fingertip direction) of the part running in the wale direction. | | | | | |



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TABLE II. Manufacturing operations requirements – Continued

| No. Requirements | Stitch Type | Seam and Stitching Type | Stitches per Inch | Thread | |
|---|---|---|---|---|---|
| | | | | Needle | Bobbin/Looper |
| Note: As an option to cutting the gloves to pattern and then joining the parts, the material to be cut shall be marked with a pattern that is 1/8 to 3/16 inch smaller than the Government pattern (i.e. margin between peripheries of the two patterns is 1/8 to 3/16 inch as measured when smaller pattern is centered on Government pattern). Two layers of the material shall then be joined by sewing along the pattern lines marked on the material while the material is being cut to the glove pattern 1/8 to 3/16 inch from the row of stitching. The stitch type, seam type, stitches per inch, and thread shall be the same as those specified in operation 3. | | | | | |
| 2. Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.2.2 shall be removed from production and replaced with non-defective and properly matched components. | | | | | |
| 3. Joining. Join front to back by stitching 1/8 to 3/16 inch from raw edge. | 301 | Ssa-1 | 8-10 | 50-3 | 50-3 |
| | or 401 | Ssa-1 | 8-10 | 50-3 | 70-2 |
| 4. Labeling. Apply label to inside of Type I gloves and the outside of Type II gloves. The label shall be centered between side seams of gloves, about 1/2 inch from unfinished edge of wrist. | | | | | |
| 5. Trimming and cleaning. Trim thread ends (unless otherwise specified) and surplus material. Remove all spots, stains, and loose ends. | | | | | |
| 6. Finishing. Turn Type I gloves right side out and form or lay off. Type II gloves shall not be turned, but shall be laid out flat. | | | | | |

* 3.8 Measurements. The finished glove shall conform to the measurements specified in Table III. The measurements shall be made as indicated on figure 1 except Type II gloves shall be measured from stitch line to top of glove and from stitch line to stitch line as applicable.



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TABLE III. Measurements of finished gloves

| Characteristic | Figure 1 | Small | Medium | Tolerance |
|---|---|---|---|---|
| Total length (inches) | A | 14 1/2 | 15 | +/- 1/2 |
| Hand width (inches) | B | 3 1/2 | 4 | +/- 1/4 |
| Width of wrist opening | C | 3 1/4 | 3 3/4 | +/- 1/8 |

3.9 Workmanship. The finished gloves shall conform to the quality of product established by this document. The occurrences of defects shall not exceed the applicable acceptable quality levels.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Certificates of compliance. When certificates for compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be preformed in accordance with ANSI/ASQC Z1.4.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.1.1 Component testing. The finished knitted fabric shall be tested as specified in table IV. The lot size shall be expressed in units of yards. The sample unit shall be 1 yard full width of the finished fabric. The number of sample units shall be as shown below. The lot shall be unacceptable if one or more sample unit fail to meet any requirement specified in Table IV. All test reports shall contain the individual values utilized in expressing the final result.

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |



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TABLE IV. Component tests

| Component | Characteristic | Requirement reference | Test Method |
|---|---|---|---|
| Cotton jersey knit | Material identification | 3.2.1 | 1/ |
| | Type of knit | 3.2.1 | 1/ |
| | Absence of optical bleaches (brighteners) | 3.2.1 | 1/ |
| | Weight per square yard | 3.2.1 | ASTM-D-3776 |
| | Wales and courses per inch | 3.2.1 | ASTM_D-3887 |
| | Bursting strength | 3.2.1 | ASTM-D-3787 |
| | Nonfibrous material | 3.2.1.1 | ASTM-D-629 |

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.3.2 End item visual examination. The end item shall be examined for the defects listed in table V. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs. The inspection level shall be I.

TABLE V. End item visual defects

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Paring | Not properly mated (i.e., not of same size) | 101 | |
| Cleanness | Spot or stain | | 201 |
| | Thread ends not trimmed | | 202 |
| | Surplus material on seams not evenly trimmed | | 203 |
| | Rancid or other objectionable odor | 102 | |
| Color | Other than bleached white | 103 | |
| Cutting | Any component not cut on straight line and not cut with length direction of component in wales direction of fabric | 104 | |
| Material and Workmanship | Any hole larger than 3/32 inch | 105 | |
| | Three or more holes | 106 | |
| | Two holes 3/32 inch or less located closer than 3 inches to each other | | 204 |
| | NOTE: Two holes 3/32 inch or less are not to be scored as a defect if the holes are 3 inches or more apart. All holes shall be measured across opening in longest dimension. | | |



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TABLE V. End item visual defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| | Run or dropped stitch | 107 | |
| | Any clearly noticeable unevenness of fabric | | 205 |
| | Needle chews | 108 | |
| | Clearly noticeable slubby yarn | | 206 |
| | Any cut, tear or burn | 109 | |
| | Any Mend | 110 | |
| Fingers and thumb | Any finger or thumb twisted, pleated, or poorly shaped | | 207 |
| Seams and Stitching | Any open seam | 111 | |
| | Repair of seam not as specified | | 208 |
| | NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur or raw edge is not caught in stitching | | |
| Raw edge (except on wrist) Type I only | More than 1/4 inch on outside of glove | | 209 |
| Seam and stitch type | Wrong seam type | | 210 |
| | Wrong stitch type | | 211 |
| Construction | Type I glove not inseam sewn | | 212 |
| | Type II glove not outseam sewn | | 213 |
| Stitches per inch | Less than six stitches per inch | 112 | |
| | Six or seven stitches per inch | | 214 |
| | More than ten stitches per inch | | 215 |
| | NOTE: To be scored only when the condition exists on the major portion of a seam. | | |
| Stitch tension | Loose tension resulting in a loosely secured seam | 113 | |
| | Tight tension resulting in clearly noticeable puckering along seam (stitches break when normal strain is applied to seam or stitching) | 114 | |
| | Loose stitching (401 stitch type) | 115 | |
| Ends of stitching | Ends of stitching produced with 301 stitch type, back tacked less than 1/4 inch | | 216 |



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TABLE V. End item visual defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Inseam | Stitching less than 1/8 or more than 3/16 inch from edge | | 217 |
| Label | Missing, incorrect; or illegible | 116 | |
| | Misplaced | | 218 |

* 4.3.3 End item dimensional examination. The end item shall be examined for the measurements listed in Table III. Amy measurement not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of gloves. The sample units of gloves. The sample unit shall be one glove. The inspection level shall be S-3.

4.3.4 Packaging examination. The fully packaged end item shall be examined for marking, materials, workmanship and content defects.

## 5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be preformed by DoD personnel, these personnel need to contact the responsible packaging maintained by the Inventory Control Points packaging activity within the Military Department of Defense Agency, or within the Military Departments System Command. Packaging data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

6.1 Intended use. Type I gloves are intended for use in handling film and other similar items. Type II gloves are intended for use as inserts for Glove Set, Chemical Protective, MIL-DTL-43976.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and size required (see 1.2).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When first article is required (see 3.1 and 6.6).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.



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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.5 Subject term (key word listing).

Clothing
Gloves
Protective clothing

6.6 First Article. When a first article is required, or shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of unity to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspections, and approval of the first article.

6.7 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army – GL
Navy – NU
Air Force – 11

Preparing activity:

DLA - CT

Project No. 8415-0276

Review activities:

Army – MD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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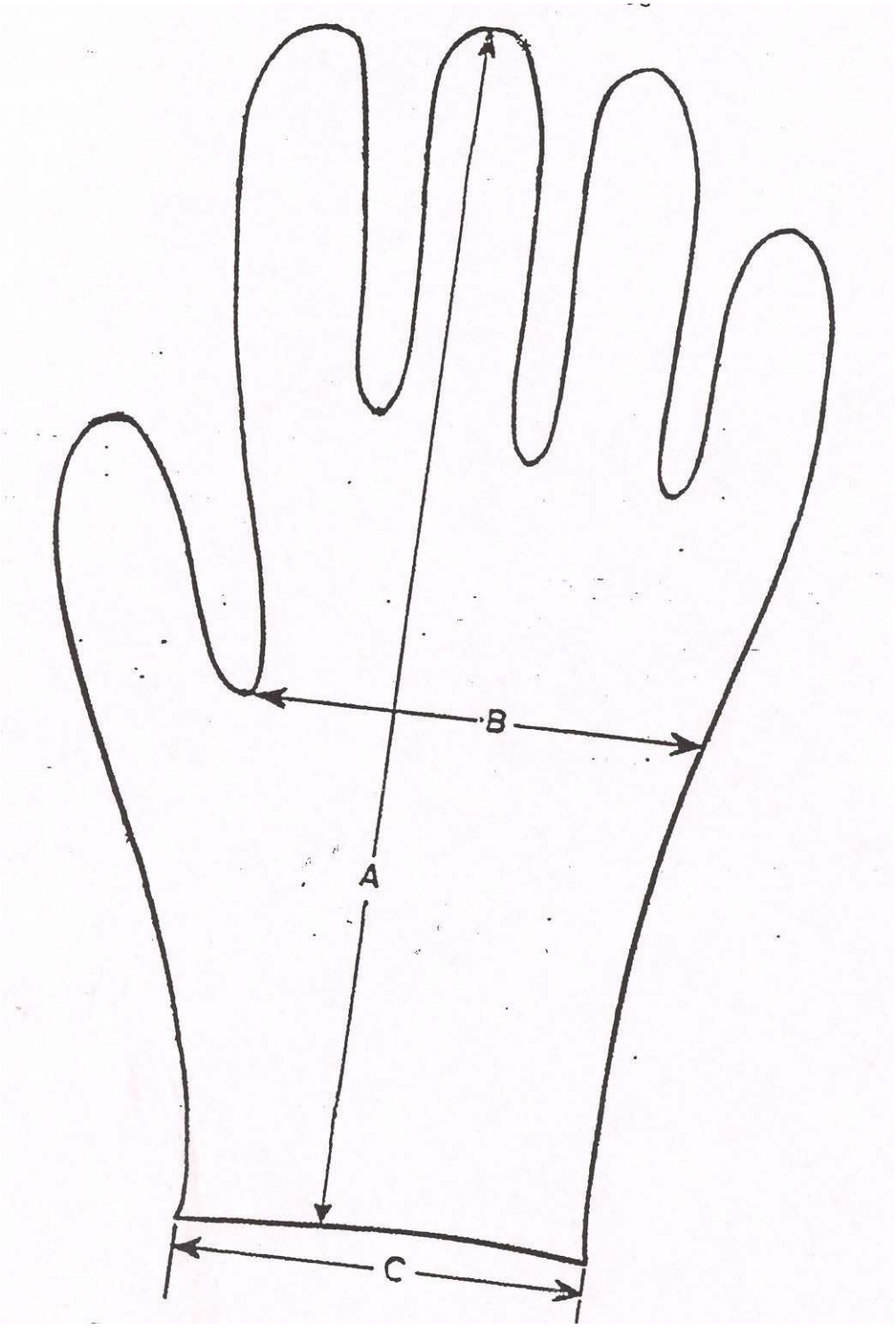



Figure 1. Gloves, Men's, Cloth, Knitted, Lightweight.



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